



Department of Computer Science
Faculty of Computing
UNIVERSITI TEKNOLOGI MALAYSIA

SUBJECT NAME: COMPUTER ORGANIZATION AND ARCHITECTURE

SUBJECT CODE: SECR 1033

SEMESTER: 2 - 2023/24

LAB TITLE: Programming 3: Interactive Usage of Link Libraries
and Comparison & Conditional Jumps

PART 1: Interactive Usage of Link Libraries
PART 2: Comparison & Conditional Jumps

INSTRUCTION: Complete the lab activities in Part 1C and Part 2C and divide loads among group member equally

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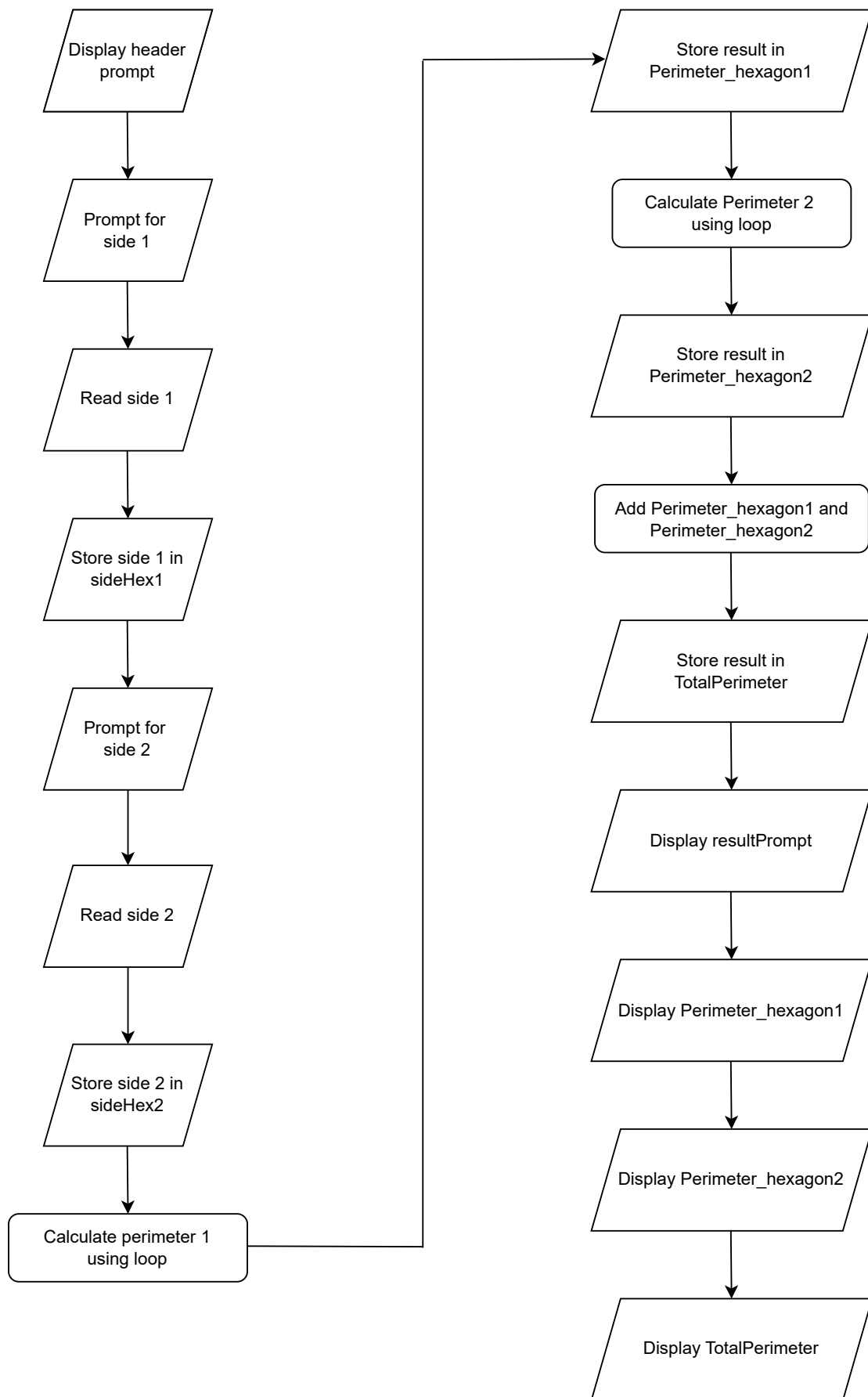
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Section : 2

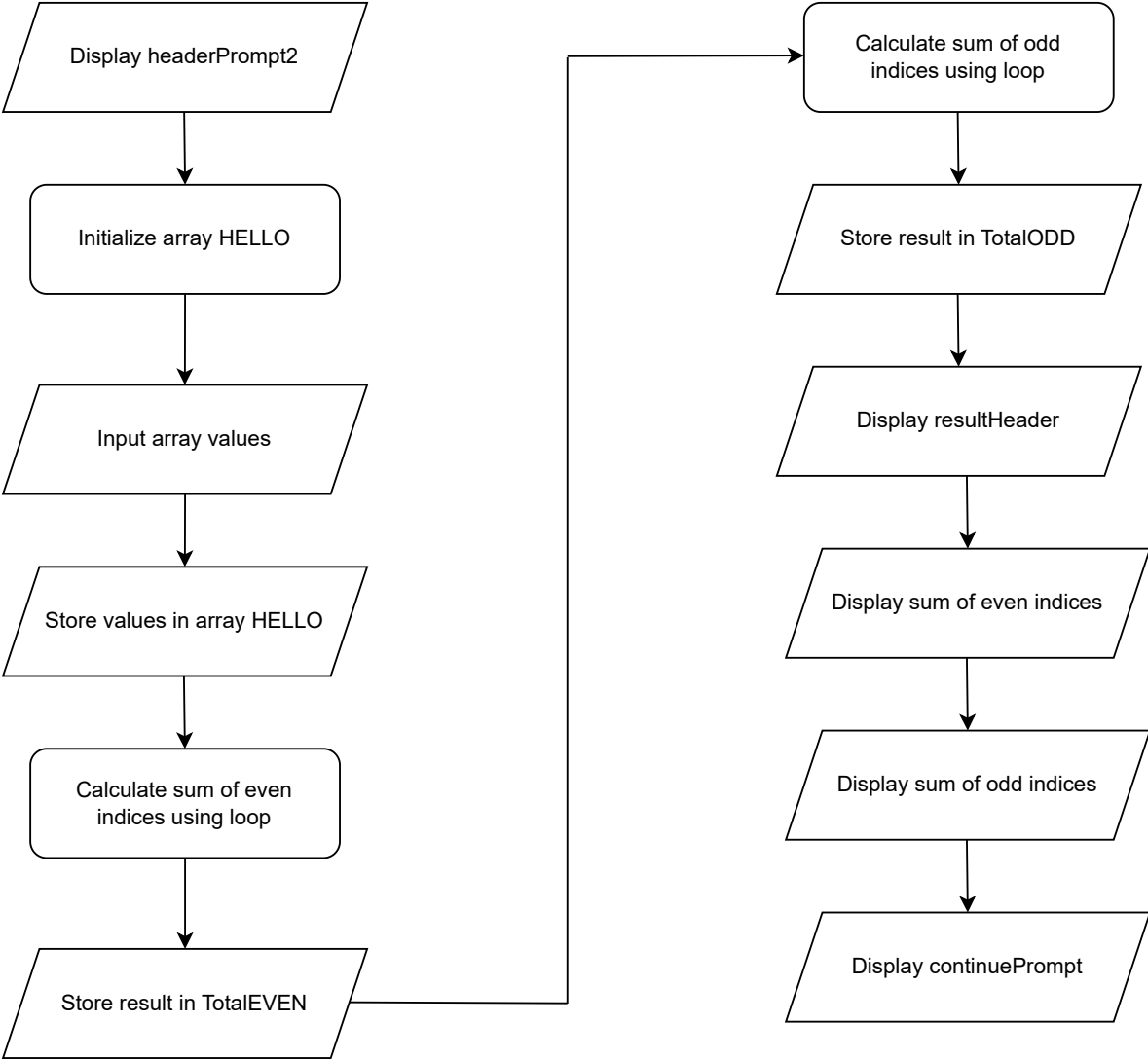
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COMMENTS:

Part1c.1



Part 1c.2



Flow Chart For Part 2C

1. **Display Main Menu:** Show the main menu options.
2. **Get User Selection:** The user selects an option (1 or 2).
3. **User Input 1?** Decision point based on user input.

3.1 Yes:

- 3.1.1 Display Title Message:** Show the title for perimeter calculation.
- 3.1.2 Input Hexagon 1 Side Length:** Get the side length for the first hexagon.
- 3.1.3 Input Hexagon 2 Side Length:** Get the side length for the second hexagon.
- 3.1.4 Calculate Perimeter Hexagon 1 and 2:** Calculate the perimeter of the first hexagon and second hexagon.
- 3.1.5 Display Perimeter of each Hexagon:** Show the perimeters of both hexagons.
- 3.1.6 Calculate Total Perimeter For Both Hexagons:** Calculate the total perimeter of the first hexagon and second hexagon.
- 3.1.7 Display Total Perimeter:** Show the total perimeters of both hexagons.
- 3.1.8 Ask to continue by typing 'y' or exit by typing 'n':** Prompt the user to continue or exit.
- 3.1.9 Continue? (y/n):** Decision point based on user input.
 - 3.1.10 Yes ('y'):** Go back to step 1.
 - 3.1.11 No ('n'):** Go to step 5.

3.2 No: Check if option 2 is selected.

4. User Input 2?

4.1 Yes:

- 4.1.1 Display Message:** Show the message for sum calculation.
- 4.1.2 Input 6 Integers:** Get six integer inputs and store them in the array.
- 4.1.3 Calculate Sum of Even Indices:** Calculate the sum of integers at even indices.
- 4.1.4 Calculate Sum of Odd Indices:** Calculate the sum of integers at odd indices.
- 4.1.5 Display Results:** Show the sum of even and odd indices.
- 4.1.6 Ask to continue by typing 'y' or exit by typing 'n':** Prompt the user to continue or exit.
- 4.1.7 Continue? (y/n):** Decision point based on user input.

4.1.8 Yes ('y'): Go back to step 1.

4.1.9No ('n'): Go to step 5.

4.2 No: Show an invalid input message and return to the main menu.

5. End Program: End the program execution.

